

V Semester

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING						
Course Code: 24CS510		Course Title: <i>Computer Networks</i>				
Total Credits: 04 L-45: T-0: P-30: SL and TW-45		SDGs addressed: 4,9,12,16,17		Course Category: PCC		Course Type: Theory with Practical
Teaching Learning Process		Total Contact Hours/Semester	Assessment Weightage and Marks			
	Lecture	45		CIE	SEE	Total
	Tutorial	-	Weightage, %	40	60	100
	Practical	30	Maximum Marks	40	60	100
	Self-learning and Term Work	45	Minimum Marks	20	25	45*
	Total	120				

Note: *For passing the student must score a Minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)

Course Pre-Requisite: Data Communication.

Course Objectives:

Sl. No.	Course Objectives
1	Introduce the various routing and congestion control algorithms.
2	Understand the various Internet protocols such as [IPV4, IPV6].
3	Analyze the different application protocols and streaming services.

Course Outcomes (COs)

CO	Course Outcomes	Highest Level of Cognitive Domain	Working Knowledge
CO1	Implement various Routing algorithms on Datagram and Virtual Circuit Network.	L3	WK2, WK3
CO2	Design of different packet scheduling and congestion control policies.	L4	WK2, WK4
CO3	Compare and contrast various protocols of Transport layer and Network layer	L2	WK3, WK4
CO4	Use of TCP service primitives for creation and connection management of applications.	L3	WK6, WK3
CO5	Analyze various transport layer and application layer concepts.	L4	WK4, WK8

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

Course Content / Syllabus:

Unit No.	Content	Lecture Hours
1	The Network layer: Network Layer Design Issues: Store-and-Forward Packet Switching, Services provided to the Transport Layer, Implementation of Connectionless Service, Implementation of	09

	Connection-Oriented Service, Comparison of Virtual-Circuit and Datagram Subnets. Routing Algorithms: The Optimality Principle, Shortest Path Routing, Flooding, Distance Vector Routing, Link State Routing, Hierarchical Routing. Self-Learning and Term Work* • Advanced Routing Protocols • IP Addressing & Subnetting • Hierarchical Routing in the Internet (Autonomous Systems & BGP).	
2	Congestion control algorithms: Approaches to congestion control, Traffic Aware Routing, Admission Control, Traffic Throttling, Load Shedding, and Quality of Service: Application Requirements, Traffic Shaping, Packet Scheduling, Admission Control, Integrated Services, and Differentiated Services. Tunneling, Packet fragmentation. Self-Learning and Term Work* • Real-world application in ISPs for handling backbone traffic. • Role of edge computing in congestion management. • QoS challenges in IoT networks with resource-constrained devices.	09
3	The Network Layer in the Internet: The IP Version 4 Protocol, IP Addresses, IP Version 6, Internet Control Protocols, Label Switching and MPLS, OSPF, BGP. Transport Layer: The Transport Service: Services provided to the upper layers. Self-Learning and Term Work* Cisco Packet Tracer/NS2/Mininet to simulate: • Distance Vector Routing (RIP). • Link State Routing (OSPF).	09
4	Berkeley Sockets, and Elements of Transport Protocols: Addressing, Connection establishment, Connection release, Error Control & Flow control Multiplexing and Crash recovery. Simple transport protocol, UDP, Remote Procedure Call, Real Time Transport Protocol. Self-Learning and Term Work* • Study of RTCP for QoS feedback. • Lightweight transport in IoT: CoAP over UDP. • Research on “NAT traversal techniques” (STUN, TURN, ICE in WebRTC).	09
5	TCP Protocol: TCP segment Header, TCP connection establishment, TCP connection release, TCP connection management, Transmission policy, Sliding window. The Application Layer: DNS: The DNS Name Space, Resource Records, Name Servers. Electronic Mail: Architecture and Services, The User agent, message formats, and message transfer, final delivery, WWW Architecture and Services Self-Learning and Term Work* • Case study: Real-world adoption of DNSSEC by ISPs. • Study SPF (Sender Policy Framework), DKIM (DomainKeys Identified Mail), DMARC. • Real-time communication: WebSockets, WebRTC for chat, video	09

	conferencing.	
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*** NOTE: Self-Learning and Term Work are evaluated only through continuous internal evaluation (CIE) and will not be assessed during the Semester End Examination (SEE).**

Text Books:

Sl. No.	Author/s	Title	Publisher Detail
1	Andrew S Tanenbaum, David J Wetherall,	Computer Networks	6th Edition, PHI/Pearson Publication, 2021.

Reference Books:

Sl. No.	Author/s	Title	Publisher Detail
1	Alberto Leon-Garcia and Indra Widjaja	Communication Networks – Fundamental Concepts and Key architectures	2nd Edition Tata McGraw-Hill, 2004.
2	William Stallings	Data and Computer Communication	8th Edition, PHI, 2007.
3	Behrouz A Forouzan	Data Communications and Networking	5th Edition, Tata McGraw Hill, 2013.
4	James F. Kurose and Keith W. Ross	Computer Networking	7th Edition, Pearson, 2017.

Web Resources:

Sl. No.	Web Links
1	https://nptel.ac.in/courses/106/105/106105081/
2	https://onlinecourses.swayam2.ac.in/cec19_cs07/preview

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	3	3	3
CO2	3	3	3	3	-	-	-	-	-	-	-	3	3	3
CO3	3	3	3	-	-	-	-	-	-	-	-	-	3	3
CO4	3	3	3	-	-	-	-	-	-	-	-	-	3	3
CO5	3	3	3	3	-	-	-	-	-	-	-	3	3	3

High – 3, Medium – 2, Low - 1

LIST OF PROGRAMS FOR PRACTICAL SESSIONS

Weeks	List of Programs	No. of Hours
PART-A: Implement the following using C++/Java/Python		
1	A. Use appropriate tool to find Host name, IP address, physical address, subnet mask, default gateway etc [hint: Ipconfig]. Explore different options available with the tool. B. On the host system, identify the TCP connections, ports on which the computer is listening, Ethernet statistics, the IP routing table, IPV4statistics, and IPV4 statistics etc. using appropriate tool [hint:	02

	netstat]. Explore different options available with the tool.	
2	Write a program to sort frames using appropriate sorting techniques	02
3	Write a program for error detecting code using CRC-CCITT (16- bits).	02
4	Write a program to find the shortest path between vertices using bellman-ford algorithm.	02
5	Using TCP/IP sockets, write a client – server program to make the client send the file name and to make the server send back the contents of the requested file if present	02
6	Write a program on datagram socket for client/server to display the messages on client side, typed at the server side.	02
7	Write a program for simple RSA algorithm to encrypt and decrypt the data.	02
8	Write a program for congestion control using leaky bucket algorithm and token bucket algorithm.	02
9	Write a program to implement random early detection (RED) congestion control algorithm.	02
PART-B: Simulation Experiments using NS2/ NS3/ NetSim or any other suitable simulation software		
10	Implement three nodes point – to – point network with duplex links between them. Set the queue size, vary the bandwidth and find the number of packets dropped.	02
11	Implement transmission of ping messages/trace route over a network topology consisting of 6 nodes and find the number of packets dropped due to congestion	02
12	Implement an Ethernet LAN using n nodes and set multiple traffic nodes and plot congestion window for different source / destination.	02
13	Implement simple ESS and with transmitting nodes in wire-less LAN by simulation and determine the performance with respect to transmission of packets	02
14	Revision	02
15	Laboratory Test	02